Xin Meng 孟昕

2027 PhD applications | 3D Medical Imaging · Multimodal Medical AI · Trustworthy Evaluation

Email: mengxin@cmu.edu.cn | mengxinlab@gmail.com | Website: mengxinlab.github.io

GitHub: github.com/mengxinlab | ORCID: 0000-0003-3451-2373

Research Profile

I am an MLIS student at the School of Health Management, China Medical University, conducting research in the Department of Clinical Epidemiology at Shengjing Hospital. I study **3D medical imaging, multimodal medical AI, and model reliability** under distribution shift through external validation, case-level failure analysis, test-time update audits, and evidence-bounded reasoning.

Education

China Medical University 2024.09 – 2027.06

Master of Library and Information Science (MLIS), in progress | Research focus: Medical AI | Shenyang, China

Jilin University 2018.08 – 2022.06

B.Mgmt. in Archival Science; B.Econ. in International Economics and Trade | Changchun, China

Research Focus and Methods

- **3D medical imaging:** Lung CT risk, longitudinal registration, patient splits, external validation, calibration, and matched architecture controls.
- **Multimodal medical AI and RAG:** Image–text fusion, text provenance, guideline retrieval, citation checks, abstention, and rule–language separation.
- **Trustworthy evaluation:** Cross-domain failures, adaptation benefit–risk, patient-clustered bootstrap, selective referral, and reproducible audits.

Accepted Papers and Conference Presentations

† denotes equal first-author contribution.

Auditing Clinical Text Leakage and Text Dominance in Multimodal Lung Nodule Assessment 2026 · Accepted

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | IEEE BIBM 2026 AIBMMD Workshop accepted paper

My role: Defined minimal, safe, leaky, and de-leaked text conditions; compared text-only, image-only, and fusion models to separate leakage-driven performance inflation from text dependence; used patient-clustered bootstrap to estimate uncertainty.

Mining Multi-Model Failure Patterns for Reliability Auditing of 3D Lung Nodule Malignancy Predictors 2026 · Accepted

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | IEEE ICDM 2026 ARRL Workshop accepted paper

My role: Built the case-level failure-mining and selective-referral pipeline across seven 3D models; compared shared errors, high-confidence mistakes, and residual risks after referral.

A Benefit–Risk Audit of Test-Time Update Budgets for Longitudinal Medical Image Registration: A GrowthMamba Case Study 2026 · Oral presentation

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Qing Chang, Yuhong Zhao | 32nd Chinese Medical Association Conference on Medical Informatics (paper in Chinese)

My role: Compared frozen inference with multiple test-time update budgets and jointly analyzed image similarity and Jacobian folding risk; under cross-domain stress, frozen inference achieved higher mean similarity than three online-update variants.

Trends and Hotspots in Deep Learning-Assisted Medical Imaging Research 2025 · Accepted

Xin Meng, Huixin Yu, Qing Chang, Yuhong Zhao | 31st Chinese Medical Association Conference on Medical Informatics (paper in Chinese)

My role: Retrieved relevant Web of Science Core Collection records from 2016–2025 and used VOSviewer to analyze keyword co-occurrence, citations, and author collaboration.

Representative Research Manuscripts

Auditing LLM-assisted scholarly reading: Replicate-session agreement in claim-span selection and evidence coding 2026 · External review

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | Computers in Human Behavior Reports

My role: Designed the reproducibility audit for LLM-assisted reading and completed the methods, experimental software, statistical analysis, and visualization; compared claim-span selection and evidence-coding agreement across repeated sessions.

Representative Research Manuscripts (continued)

Auditable Decision-Invariant Retrieval-Augmented Lung-RADS Report Realization: A Verified-Subset Technical Evaluation 2026 · Under review

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | IEEE JBHI

My role: Designed a pipeline where deterministic rules own Lung-RADS categorization and retrieval and the LLM only realizes fixed evidence and wording; audited citation validity, abstention, text leakage, and unsupported generation.

Source-calibrated CT deep learning for lung nodule malignancy risk: external validation and matched-control analysis across public cohorts 2026 · Manuscript in preparation

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang

My role: Designed matched CNN, Mamba, and attention controls; performed external validation, calibration, and fixed-operating-point evaluation on LNDb and LIDC-IDRI. Additional Mamba layers or geometric metadata showed no clear gain.

When "Not Documented" Is Not Clean: A Lineage-Aware Audit and Controlled Stress Test of CT Pretraining 2026 · Conference review

Xin Meng, Wenhao Zhu

My role: Audited direct and transitive relationships between CT pretraining corpora and evaluation sets for each configuration; tested representation contamination under controlled changes in pretraining scale and random seed, retaining clear limits on negative findings.

DiTA: Gradient-Level Analysis of Dual-Teacher Distillation for Source-Free State-Space Lung Nodule Malignancy Classification 2026 · Manuscript in preparation

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang

My role: Implemented SegMamba student and CNN/Transformer dual-teacher experiments; analyzed feature- and logit-level gradient conflicts and completed LNDb transfer, test-time adaptation, and compute audits. No stable test-time update benefit emerged.

Other First- and Co-First-Author Manuscripts (Selected)

Cross-omics evaluation of genetically prioritized druggable-gene hypotheses in sarcoidosis: a Mendelian randomization study 2026 · External review

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | Orphanet Journal of Rare Diseases

Government suicide-prevention policies share broad agendas but rarely specify responsibility 2026 · External review

Xiaopeng Li†, **Xin Meng**†, Huixin Yu†, Yuhong Zhao, Qing Chang

General-Purpose Vision-Language Models versus Task-Specific Models for Pulmonary Nodule Malignancy Prediction 2026 · Revising for resubmission

Xin Meng†, Xiaopeng Li†, Huixin Yu†, Yuhong Zhao, Qing Chang | KJR (revising after external review)

Other research covers source-bounded reasoning, longitudinal nodule growth assessment, nutrition informatics, and public-health data analysis. The full manuscript list is at mengxinlab.github.io.

Research and Data Experience

National Health Commission Health Development Research Center 2025.11 – 2026.02

Remote research assistant, Prof. Wudong Guo's group; harmonized multi-province off-label drug-use consensus documents and group standards into a structured comparison table

Shanghai Zhizhongda Information Technology Co., Ltd. 2024.05 – 2024.08

Data analyst; used SQL/Tableau to analyze quotations, orders, and competitor prices in support of pricing rules

Gold Hongye Paper Group (APP / Qingfeng) 2023.04 – 2024.05

Data analysis management trainee; analyzed prices and promotions across five regions and 100+ core SKUs

Technical Skills and Academic Exchange

- **Programming and modeling:** Python, PyTorch, pandas, SQL, Git; 3D CNNs, Mamba, Transformers, knowledge distillation, multimodal fusion, and RAG.
- **Medical imaging and evaluation:** CT preprocessing, HU normalization, voxel spacing, patient-level splitting, external-cohort validation, calibration, subgroup and failure-mode analysis.
- **Academic exchange and credentials:** *Beyond the Leaderboard*, a medical-AI science-communication film, screened at the 32nd Chinese Medical Association Conference on Medical Informatics (2026); Science Communication Pioneer, 31st conference competition (2025); Information Systems Supervisor, National Intermediate Professional Certificate.
- **Languages:** IELTS 6.5 (Reading 7.5, 2024); CET-6 502.